EPISTEMIC-AWARE DRO · 72 US ETFs · 2010–2024

Three pre-registered attempts.

The holdout is still untouched.

A research-engineering portfolio project — and what I learned trying to make uncertainty pay.

Working paper · current-listed ETF universe · not survivorship-free

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THE QUESTION

Can uncertainty improve a DRO portfolio?

A mean-CVaR optimizer can shrink bets when the forecast is uncertain. The hypothesis: an ML uncertainty layer should improve the tail-risk / return trade-off, net of costs.

I tested it three ways. v1 was closed after a reduced test. v2 and v3 were stopped before the holdout by their information gates.

v1 epistemic uncertainty ⟶ H1/H3 fail, H2 non-inferiority pass

v2 conformal quantiles ⟶ gate 65.4%

v3 conditional tail risk ⟶ gate 43.0%

The mean signal was not there.

Ridge B0, 2010–2024: pooled OOS −0.019 R^2; mean rank IC 0.013, $t = 0.56$.

A date-blocked information audit found no feature surviving the 80% rule. The reduced DRO did not separate from the zero-sigma comparator (B5).

Strategy	Sharpe	Max DD	CVaR95
M1 reduced	0.816	−24.2%	−8.3%
B5 zero-sigma DRO	0.798	−24.3%	−8.4%
1/N	0.673	−26.6%	−9.8%

B5 uses zero sigma and is identical to B3 CVaR-only; the DRO-vs-no-DRO comparison is uninformative.

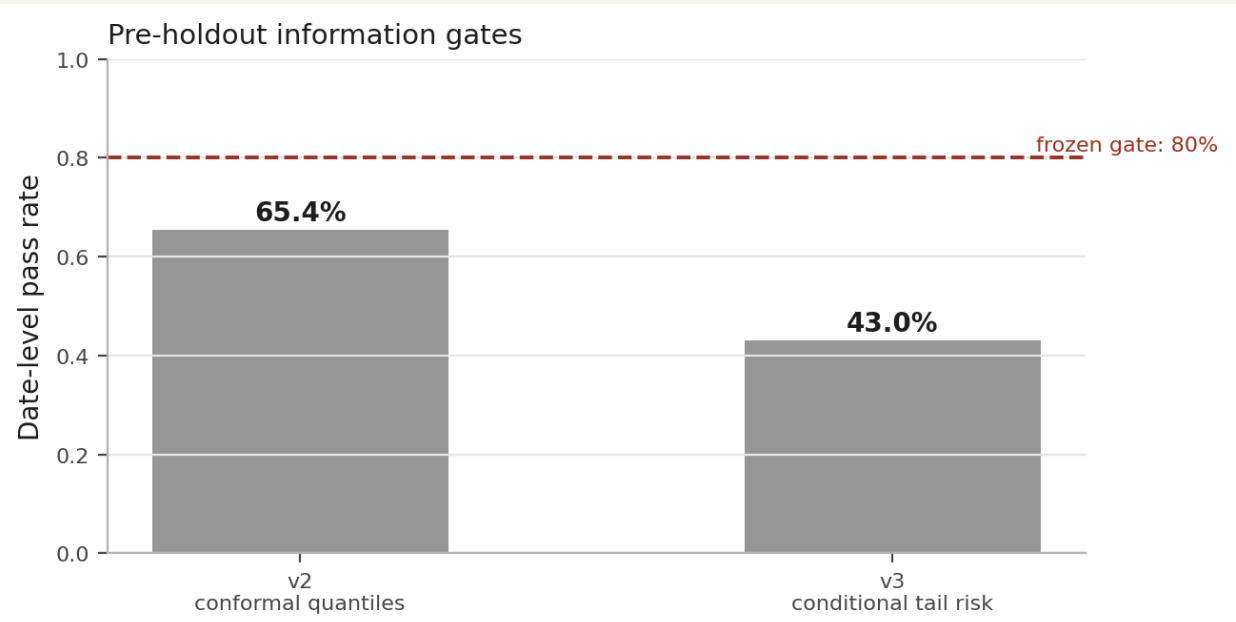
H1 / H3 fail · H2 non-inferiority pass

No alpha claim. The DRO/CVaR optimizer beat 1/N; the uncertainty layer did not separate.

Calibration worked. Information did not.

LightGBM quantile regression at q05/q50/q95, three seeds, split-conformal calibration. Coverage was 89.9% on validation and 85.9% on development test.

The frozen gate asked whether the conformal sigma carries stable date-level information. It passed 65.4% against an 80% bar.

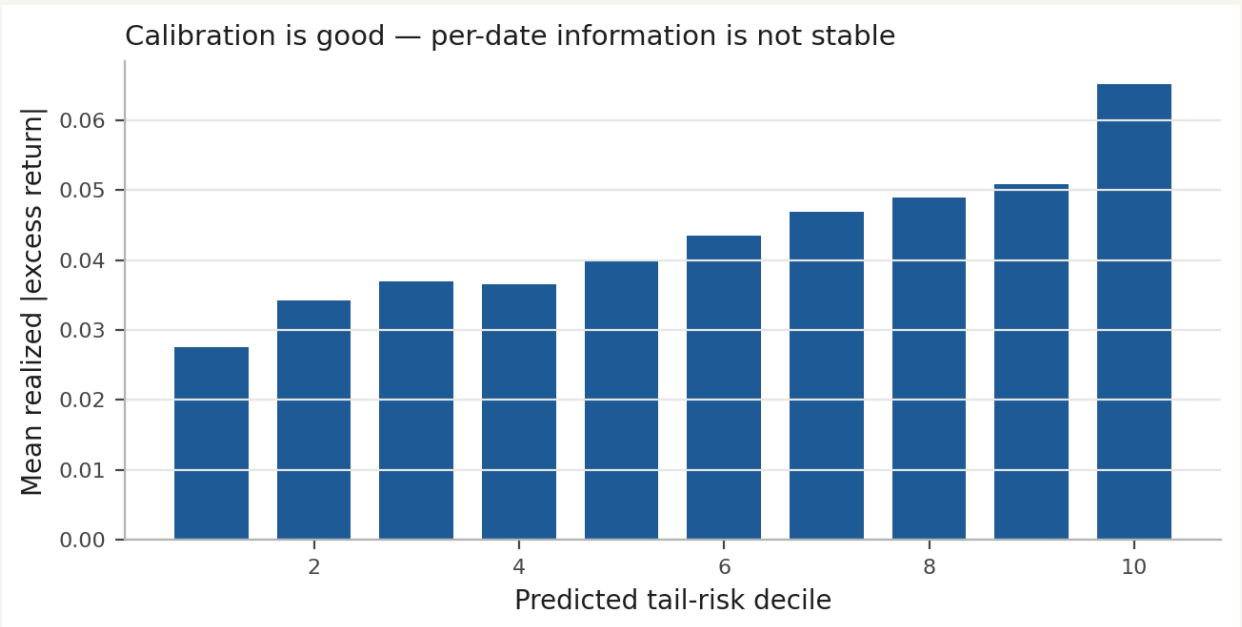


Stopped before the 2025–2026 holdout.

A calibrated forecast can still be uninformative.

LightGBM L1 forecast of next-month absolute excess return. Mean sigma 4.29% vs realized 4.30%. Correlation 0.28. Top-minus-bottom decile spread 3.76%.

It passed 43.0% of date-level information tests and underperformed trailing volatility on development.



Stopped before the holdout. Still untouched.

I tested the gate in both directions.

The 80% date-level MI gate had zero power at 72 assets — even at a 20% R^2 signal. The iid pooled gate detected it in every seed, but it was too permissive.

Signal	80% MI gate	Pooled gate
3.8% R^2	0%	100%
20% R^2	0%	100%

Synthetic, 72 assets, 179 dates, linear signal, 2 seeds.

A block-aware null cut 20/21 apparent passes to 3/21. The v2 and v3 sigma did not survive.

Too strict is not the same as too permissive. The stops were about stability.

Post-hoc methods audit: the block-aware/effect-size checks support the stops ex post; they do not prove no signal exists.

Calibration ≠ information.

Gates > narratives.

The mean signal is the binding constraint. A better optimizer cannot rescue a weak forecast.

Stability is the hard part. Average calibration and a decile spread are not enough.

Preserve the holdout. One completed result, two pre-holdout stops, zero holdout touches. The apparatus is the result.

Effect size decides. The survivors' IC intervals included zero; trailing volatility dominated the conditional sigma. No v4 on this data.

Stack: purged walk-forward · Ridge · LightGBM quantiles · conformal · cvxpy · CI + verifier.

Paper + code: github.com/kira-ml/epistemic-dro

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